BUCORIDE – WEB SYSTEM DEVELOPMENT REQUIREMENTS

Project: BUCORIDE Ride-Hailing and Parcel Delivery Platform.

Theme : Yellow and it's fadings , dark grey .Live background.

1. SYSTEM OVERVIEW

BUCORIDE is a web-based ride-hailing and parcel delivery platform connecting drivers, motorbike riders and customers. The system should have three main interfaces: Driver Dashboard, Customer Dashboard and Administration Dashboard. It must support real-time GPS tracking, automatic driver matching, ride requests, parcel delivery, monthly subscriptions and automated fare calculations.

2. DRIVER/RIDER REGISTRATION

Drivers should create accounts by entering their full name, phone number, email address and password. They should then register their vehicles by providing the vehicle model and registration number (number plate). They should select whether they are registering a CAR or MOTORCYCLE and choose their service category, either PRIVATE HIRE/CHAUFFEUR or PSV. Drivers should be able to manage their registered vehicles and accounts from their dashboards.

3. DRIVER SUBSCRIPTION SYSTEM

BUCORIDE will operate on monthly subscriptions, with motorcycles charged KSh 249 and cars KSh 549. Every newly registered driver receives the first month free. However, drivers must activate their free trial subscription before becoming eligible to receive actual customer bookings. Subscriptions should expire automatically after 30 days unless renewed. The system should display subscription status, expiry dates, payment history and send renewal reminders. Expired subscriptions should automatically lose the ability to accept new bookings.

4. DRIVER RIDE OPPORTUNITY NOTIFICATIONS

The system should periodically send drivers notifications showing generated ride opportunities with random pickup locations, destinations, distances and estimated amounts. The purpose is to demonstrate potential business opportunities and encourage drivers to register and remain active on BUCORIDE. These must be clearly identified as sample opportunities rather than genuine customer bookings. The driver dashboard should display the total number of opportunities received, missed opportunities and their combined estimated value. Actual customer bookings must be recorded separately.

5. GOOGLE MAPS AND GPS INTEGRATION

BUCORIDE must integrate the latest suitable Google Maps Platform APIs, including mapping, location detection, route calculation and distance estimation. The system should automatically detect the customer's current location after permission is granted and record their general location as their home town. Customers should also be able to change their location manually. Drivers should share their live GPS locations whenever online. The system must calculate actual road distances rather than straight-line distances.

6. CUSTOMER REGISTRATION

Customers should register using their full name, phone number, email address and password. The system should identify their device and request location permission during registration. Each customer should have access to ride requests, parcel delivery, booking history, ongoing trips and parcel tracking through their dashboard.

7. CUSTOMER FREE RIDE POLICY

Every newly registered customer should receive ONE free ride worth a maximum of KSh 100, with the cost covered entirely by BUCORIDE management. The offer should be limited to one ride per person and one ride per device. The system must implement strong anti-abuse mechanisms using device fingerprinting, phone verification and other appropriate identification methods to prevent customers from creating multiple accounts on the same device to claim additional free rides. Any fare exceeding KSh 100 should be paid by the customer.

8. PASSENGER RIDE REQUESTS

Customers should be able to request rides by selecting their pickup location and entering their desired destination. Google Maps should automatically calculate the distance and generate an estimated fare, which must be displayed before submission. Customers and drivers should have the ability to negotiate fares. Customers may propose an alternative amount, while drivers can accept, reject or counteroffer. The final fare must be agreed upon by both parties before the journey begins.

9. AUTOMATIC DRIVER MATCHING

Once a customer submits a ride request, the system should identify the nearest available and eligible driver based on GPS location, vehicle category and subscription status. The selected driver receives an instant notification with the trip details and can ACCEPT or REJECT. If accepted, the booking immediately disappears from other drivers' available requests. If rejected or unanswered within a specified period, the system automatically forwards the request to another nearby driver until one accepts or the customer cancels. The system must prevent multiple drivers from accepting the same booking.

10. DRIVER AND CUSTOMER INFORMATION

After a driver accepts a ride, the customer should see the driver's full name, phone number, vehicle model, registration number and current location. The driver should see the customer's name, phone number, pickup location, destination, estimated distance and proposed fare. Both parties should have sufficient information to communicate and coordinate the pickup.

11. PARCEL DELIVERY SERVICES

Customers should be able to request parcel delivery by entering the pickup location, destination, sender's name and phone number, receiver's name and phone number, and parcel description. The system should calculate the distance and generate an estimated delivery charge before forwarding the request to the nearest eligible driver or motorcycle rider. The driver should receive a notification and either accept or reject the request.

12. REAL-TIME PARCEL TRACKING

Once a rider accepts a parcel delivery, the system should activate live GPS tracking. Both sender and receiver should be able to follow the parcel's movement through a live map showing the rider's current location, pickup point, destination and estimated delivery time. The system should automatically update the rider's location and notify both parties when the parcel is collected, is in transit, approaches its destination and is successfully delivered. Tracking should also be accessible through a secure tracking link.

13. FARE CALCULATION AND PRICING

BUCORIDE should have an automated and configurable pricing engine based on actual road distance. For benchmarking, the developer should research the latest Bolt Kenya fares and use them as a reference when establishing BUCORIDE's initial pricing. Pricing should not be permanently hardcoded because fares vary by location, vehicle category and other factors. The administrator should be able to configure motorcycle and car rates per kilometre, base fares, minimum fares, additional charges and promotional discounts. The final fare should remain negotiable between the driver and customer.

14. ESTIMATED JOURNEY TIME

The system should automatically calculate estimated journey time using an initial cruising speed of 60 KM/H. The formula should be: Distance ÷ 60 × 60 minutes. For example, 10 KM should take approximately 10 minutes, 30 KM approximately 30 minutes and 60 KM approximately one hour. Where available, Google Maps traffic information and road conditions should also be considered to improve arrival estimates.

15. DRIVER DASHBOARD

The driver dashboard should display personal details, registered vehicles, subscription status, expiry dates, online/offline status, current location, available requests, accepted rides, completed rides, missed sample opportunities, estimated value of missed opportunities, parcel requests, completed deliveries, payment history and notifications. Drivers should have the ability to switch

between ONLINE and OFFLINE. Only eligible online drivers should receive actual customer booking notifications.

16. CUSTOMER DASHBOARD

The customer dashboard should display personal details, current location, ride request option, parcel delivery option, free ride eligibility, free ride redemption status, ongoing bookings, previous rides, parcel tracking, booking history and notifications. The interface should be simple and easy to navigate, particularly on smartphones.

17. ADMINISTRATION DASHBOARD

BUCORIDE management should have a powerful administration dashboard with complete control over the platform. Administrators should be able to view registered drivers, customers and vehicles; approve, suspend or deactivate accounts; monitor subscriptions and payments; configure subscription charges and vehicle pricing; monitor rides and parcel deliveries; view active driver locations where location sharing is enabled; manage free ride promotions; identify duplicate device registrations; send notifications; generate reports; monitor revenue and manage complaints. Management should also control the geographical areas in which BUCORIDE operates.

18. NOTIFICATION SYSTEM

The platform should support automated real-time notifications through website notifications, push notifications, SMS and email where configured. Drivers should receive notifications about new rides, parcel requests, sample opportunities, subscription expiry and payment confirmations. Customers should receive notifications about booking confirmation, driver acceptance, driver arrival, parcel collection, parcel movement and successful delivery.

19. SECURITY AND RELIABILITY

The system must implement secure authentication, encrypted passwords, secure payment verification, device-based promotional abuse prevention, GPS permission management, protected customer and driver information, database backups and activity logs. Location information should only be shared with authorized parties for legitimate booking and tracking purposes.

20. DESIGN AND USER EXPERIENCE

BUCORIDE should have a modern, professional and fully responsive interface optimized for smartphones, tablets and computers. Navigation should be simple, loading speeds fast and the booking process straightforward. The system should remain usable even on relatively slow internet connections.

21. FINAL DEVELOPMENT EXPECTATIONS

BUCORIDE must be developed as a fully functional ride-hailing and parcel delivery platform, not merely a website with registration forms and static dashboards. All features must be connected to a working backend and database, with real-time GPS, automatic driver matching, Google Maps integration, subscription management, payment processing, notifications and live parcel tracking. The system must undergo thorough testing before deployment.

EXPECTED FINAL PRODUCT: A powerful, scalable and reliable BUCORIDE transportation ecosystem where customers can request rides and parcel deliveries, drivers can register vehicles and receive bookings, and management can control the entire network from one central administration dashboard, with the capacity to expand across Kenya and eventually into other markets.